

TrES-5b

R-1949

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_160623 · created 2026-08-06T15:24:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457562.8634 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.054
Transit depth (Rp/Rs)^2	2.97 +/- 1.87 [%]
Orbital Inclination (inc)	80.34 +/- 2.86
Airmass coefficient 1 (a1)	0.044 +/- 0.024
Airmass coefficient 2 (a2)	3.0 +/- 1.45
Scatter in the residuals of the lightcurve fit is	59.78 %
Best Comparison Star	None
Optimal Aperture	5.42
Optimal Annulus	13.09
Transit Duration (day)	0.038 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.172 $\pm$ 0.054 vs 0.142 (deviation 0.41 σ)
Duration fit vs published	0.038 d vs 0.0758 d (deviation 0.92 σ)
Mid-time O-C	-3.6 min
Depth SNR	2.3
Residual scatter	59.78 %

Light curve

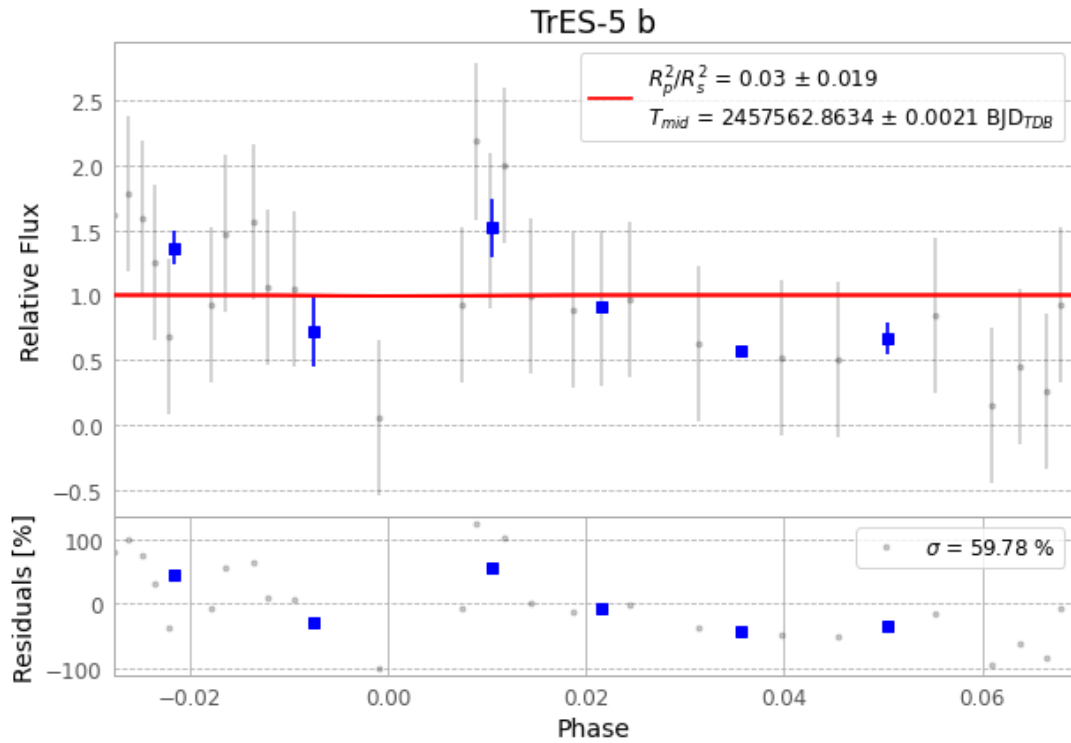


Figure 1 — FinalLightCurve_TrES-5 b_2016-06-23.png (SHA-256 c10c6f608f13c880...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [225, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9572785d25da455f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

73 FITS frames (48 MB), TRES-5160623074212.FITS → TRES-5160623111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-160623.log (c82df93e17a9...), FinalLightCurve_TrES-5 b_2016-06-23.png (c10c6f608f13...), FinalLightCurve_TrES-5 b_2016-06-23.csv (fe08df572c4c...)

Compute resources

Wall time 286.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 15:24Z.